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3/29/2026
COP 3035C-042
Homework 4

I created utils.py which is a module containing several functions and I created homework4.py to test the functions in utils.py.

## utils.py

This is the module containing the functions for the assignment.

```
# COP 3035C
# Homework 4
# Joseph Leone
# utils.py

def letter_grade(num_grade, table):
    """ takes a numerical grade and a reference table and returns a letter
grade """
    letter_grade = None
    for lower_bound, upper_bound, letter in table:
        if num_grade >= lower_bound and num_grade <= upper_bound:
            letter_grade = letter
            break
    return letter_grade

def student_loan_payment(principal, annual_interest_rate, loan_term_years):
    """ takes the principal, annual interest rate, and number of years for
the loan
        and returns the monthly loan payment """
    monthly_interest_rate = annual_interest_rate / 12
    payment_count = loan_term_years * 12
    if monthly_interest_rate > 0:
        numerator = principal * monthly_interest_rate
        denominator = 1 - (1 + monthly_interest_rate)**(-payment_count)
        monthly_payment = numerator / denominator
    elif monthly_interest_rate == 0:
        monthly_payment = principal / (payment_count)
    else:
        monthly_payment = None
    return round(monthly_payment, 2)
```

```python
def word_count(essay):
    """ takes a string of text and returns the word count and character
count """
    word_count = 0
    char_count = 0
    inside_word = False
    for char in essay:
        if char != " ":
            char_count = char_count + 1
        if inside_word == True:
            if char.isalnum() == False:
                inside_word = False
        else:
            if char.isalnum() == True:
                word_count = word_count + 1
                inside_word = True
    return word_count, char_count

def convert_temperature(degrees, unit):
    """ takes the temperature in C or F and converts it to the other unit
"""
    if unit == 'C':
        temperature = degrees * (9 / 5) + 32
    elif unit == 'F':
        temperature = (5 / 9) * (degrees - 32)
    else:
        return "Invalid unit! Use 'C' for Celsius or 'F' for Fahrenheit."
    return temperature

def height_weight_metric(feet, inches, weight_pounds):
    """ takes height in two parameters, feet and inches and takes weight
in pounds
        and returns the height in cm and weight in kg """
    height_inches = (feet * 12) + inches
    height_cm = round(height_inches * 2.54, 1)
    weight_kg = round(weight_pounds * 0.45359237, 1)
    return (height_cm, weight_kg)
```

homework4.py

This script tests the functions from the module utils.py.

```
# COP 3035C
# Homework 4
# Joseph Leone
# homework4.py

import utils

def main():
    # testing utils.letter_grade
    table = [
        (90,100,"A"),(87,89,"A-"),(83,86,"B+"),
        (80,82,"B"),(77,79,"B-"),(73,76,"C+"),
        (70,72,"C"),(67,69,"C-"),(63,66,"D+"),
        (60,62,"D"),(51,59,"D-"),(0,50,"F")
    ]
    test_inputs = [100, 25, 75]
    print("-"*64)
    print(f"If table = {table}")
    for numerical_grade in test_inputs:
        print(f"For numerical grade = {numerical_grade}")
        letter_grade = utils.letter_grade(numerical_grade, table)
        print(f"letter_grade outputs {letter_grade}")
    print("-"*64)

    # testing utils.student_loan_payment
    test_inputs = [
        (30000, 0.05, 10),
        (50000, 0.04, 15),
        (10000, 0.0, 5)
    ]
    for principal, yearly_interest_rate, loan_term_years in test_inputs:
        print(f"If principal = {principal}, yearly_interest_rate =
{yearly_interest_rate} and loan_term_years = {loan_term_years}")
        monthly_payment = utils.student_loan_payment(principal,
yearly_interest_rate, loan_term_years)
        print(f"student_loan_payment outputs {monthly_payment}")
    print("-"*64)

    # testing utils.word_count
```

```
    test_essay = "Python is a powerful programming language that is widely used in various fields."
    print(f"For text input: \"{test_essay}\"")
    words, characters = utils.word_count(test_essay)
    print(f"word_count outputs word_count = {words} and character_count = {characters}")
    print("-"*64)

    # testing utils.convert_temperature
    test_inputs = [
        (0, "C"),
        (100, "C"),
        (32, "F"),
        (212, "F"),
        (50, "X")
    ]
    for degrees, unit in test_inputs:
        print(f"If degrees = {degrees} and unit = {unit}")
        temperature = utils.convert_temperature(degrees, unit)
        print(f"convert_temperature outputs {temperature}")
    print("-"*64)

    # testing utils.height_weight_metric
    test_inputs = [
        (5, 4, 120),
        (0, 72, 205),
    ]
    for height_feet, height_inches, weight_pounds in test_inputs:
        print(f"height_feet = {height_feet}, height_inches = {height_inches}, and weight_pounds = {weight_pounds}")
        height_cm, weight_kg = utils.height_weight_metric(height_feet, height_inches, weight_pounds)
        print(f"utils.height_weight_metric outputs height_cm = {height_cm} and weight_kg = {weight_kg}")
    print("-"*64)

if __name__ == "__main__":
    main()
```

homework4.py Output

This is the output after running the tests on the functions from the module utils.py.

```
----------------------------------------------------------------
If table = [(90, 100, 'A'), (87, 89, 'A-'), (83, 86, 'B+'), (80, 82, 'B'), (77, 79,
'B-'), (73, 76, 'C+'), (70, 72, 'C'), (67, 69, 'C-'), (63, 66, 'D+'), (60, 62, 'D'),
 (51, 59, 'D-'), (0, 50, 'F')]
For numerical grade = 100
letter_grade outputs A
For numerical grade = 25
letter_grade outputs F
For numerical grade = 75
letter_grade outputs C+
----------------------------------------------------------------
If principal = 30000, yearly_interest_rate = 0.05 and loan_term_years = 10
student_loan_payment outputs 318.2
If principal = 50000, yearly_interest_rate = 0.04 and loan_term_years = 15
student_loan_payment outputs 369.84
If principal = 10000, yearly_interest_rate = 0.0 and loan_term_years = 5
student_loan_payment outputs 166.67
----------------------------------------------------------------
For text input: "Python is a powerful programming language that is widely used in va
rious fields."
word_count outputs word_count = 13 and character_count = 68
----------------------------------------------------------------
If degrees = 0 and unit = C
convert_temperature outputs 32.0
If degrees = 100 and unit = C
convert_temperature outputs 212.0
If degrees = 32 and unit = F
convert_temperature outputs 0.0
If degrees = 212 and unit = F
convert_temperature outputs 100.0
If degrees = 50 and unit = X
convert_temperature outputs Invalid unit! Use 'C' for Celsius or 'F' for Fahrenheit.
----------------------------------------------------------------
height_feet = 5, height_inches = 4, and weight_pounds = 120
utils.height_weight_metric outputs height_cm = 162.6 and weight_kg = 54.4
height_feet = 0, height_inches = 72, and weight_pounds = 205
utils.height_weight_metric outputs height_cm = 182.9 and weight_kg = 93.0
----------------------------------------------------------------
```